

a price (PV) of EUR95.72 per EUR100 principal. What is the YTM on this bond?

Solution:

1.10 percent

Use Equation 5 to solve for r given a PV of 95.72, FV of 100, and $t = 4$, we get:

$$\text{EUR}95.72 = \text{EUR}100 / (1 + r)^4; r = 1.10\%,$$

or

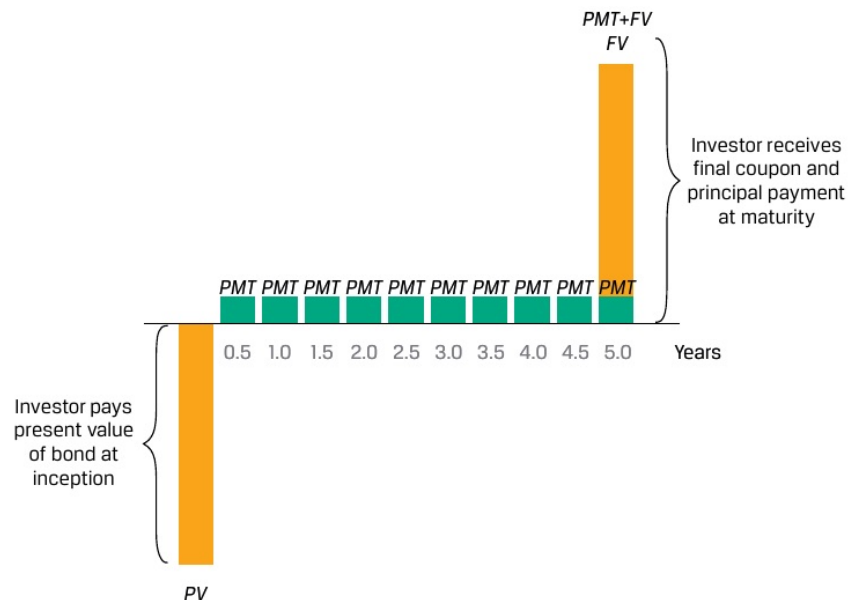
$$1.10\% = \text{RATE}(4, 0, -95.72, 100, 0, 0.1).$$

In the case of the Republic of India discount bond, a higher interest rate will reduce the bond's price. However, in contrast to the accreting price of a discount bond, the premium bond's price will decline (or amortize) over time to reach the EUR100 future value at maturity.

Coupon Instrument

A periodic cash flow pattern for fixed-income interest payments with principal repaid at maturity is shown in Exhibit 2. In this case, all the periodic cash flow payments are identical and occur on a semiannual basis.

Exhibit 2: Coupon Bond Cash Flows



Pricing a coupon bond extends the single cash flow calculation for a discount bond to a general formula for calculating a bond's price (PV) given the market discount rate on a coupon date, as follows:

$$\begin{aligned} &PV(\text{Coupon Bond}) \\ &= PMT_1 / (1 + r)^1 + PMT_2 / (1 + r)^2 + \dots + (PMT_N + FV_N) / (1 + r)^N. \end{aligned} \quad (6)$$